

Week 2 Online Assignment - Solutions

Ex 1 Twitter Posts

Suppose the observational units for some data are Twitter posts. Say whether each of the following is a variable or not. If it is a variable, give its type. If it is categorical, also say whether it is binary.

Item	Variable?	Type	Why
a) Number of words in the post	Yes	Quantitative	It counts the words, and each post has its own value.
b) Number of days since the post	Yes	Quantitative	It measures an amount of time for each post.
c) Number of retweets of the post	Yes	Quantitative	It counts the retweets of each post.
d) Total number of posts by a person	No	–	This describes a person, not a single post. It would be a variable if the observational units were Twitter users.
e) Average number of words for all posts by a certain user on a certain day	No	–	This is one summary of many posts, not a value recorded for each post.
f) Whether or not a post has a reply to it	Yes	Categorical, binary	Each post is Yes or No.
g) First word in the post	Yes	Categorical, not binary	The first word is a category, and there are many possible words.
h) Whether or not the post uses any capital letters	Yes	Categorical, binary	Each post is Yes or No.

Ex 2 Sales Transactions

Sales Transactions: July 14

Cust ID	Region	Payment	Transaction Code	Source	Amount	Product	Time Of Day
10001	East	Paypal	93816545	Web	\$20.19	DVD	22:19
10002	West	Credit	74083490	Web	\$17.85	DVD	13:27
10003	North	Credit	64942368	Web	\$23.98	DVD	14:27
10004	West	Paypal	70560957	Email	\$23.51	Book	15:38
10005	South	Credit	35208817	Web	\$15.33	Book	15:21
10006	West	Paypal	20978903	Email	\$17.30	DVD	13:11
10007	East	Credit	80103311	Web	\$177.72	Book	21:59
10008	West	Credit	14132683	Web	\$21.76	Book	4:04
10009	West	Paypal	40128225	Web	\$15.92	DVD	19:35
10010	South	Paypal	49073721	Web	\$23.39	DVD	13:26

- a) **Identify the observational units.** The observational units are the **sales transactions** on July 14. Each row is one transaction.
- b) **What are the variables?** The variables are the columns: Cust ID, Region, Payment, Transaction Code, Source, Amount, Product, and Time Of Day.
- c) **Name each variable and give its type.**

Variable	Type	Binary?
Cust ID	Categorical. The number only identifies the customer.	No
Region	Categorical (East, West, North, South)	No
Payment	Categorical (Paypal, Credit)	Yes
Transaction Code	Categorical. The number only identifies the transaction.	No
Source	Categorical (Web, Email)	Yes
Amount	Quantitative. It measures money spent.	–
Product	Categorical (DVD, Book)	Yes
Time Of Day	Quantitative. It measures the time of the transaction.	–

Cust ID and Transaction Code use digits, but they do not measure an amount, so they are categorical, like ZIP codes and SKUs.

Payment, Source, and Product each have only two categories in this data, so they are binary here. If the store also took other payment types or sold other products, they would not be binary.

Ex 3 Parameter vs Statistic

1) **Time to return an item.** We want to study how long customers wait, on average, to return something. We record 50 customers' time to return an item (the number of days from buying something to returning it) and get an average of 7.5 days.

a) **Population:** All customers who return an item.

b) **Sample:** The 50 customers whose return times were recorded.

c) **Parameter:** The mean number of days to return an item for all customers who return something, μ . Its value is unknown.

d) **Statistic:** The mean number of days for the 50 sampled customers, $\bar{x} = 7.5$ days.

2) **Coke or Pepsi.** We want to study whether people prefer Coke or Pepsi. We collect 123 responses that say Coke, 89 that say Pepsi, and 12 that say neither.

a) **Population:** All people whose soda preference we want to know.

b) **Sample:** The $123 + 89 + 12 = 224$ people who responded.

c) **Parameter:** The proportion of all people who prefer Coke, p . Its value is unknown. (The proportion of all people who prefer Pepsi, or who prefer neither, are also parameters of interest.)

d) **Statistic:** The proportion of the sample who prefer Coke:

$$\hat{p} = \frac{123}{224} \approx 0.549 \text{ (54.9\%)}$$

For Pepsi, $\hat{p} = 89/224 \approx 0.397$ (39.7%). For neither, $\hat{p} = 12/224 \approx 0.054$ (5.4%).

Ex 4 Retail Variables

Below are some terms related to retail items you might keep track of in a store.

a) **What are the observational units?** The observational units are the **retail items (products)** in the store.

b) **Identify the type of each variable.**

Variable	Type	Binary?
Product tree (categories, family, product type)	Categorical	No
Reference number	Categorical. The number only identifies the item, like an SKU.	No
Product name	Categorical	No
Product status (new or not)	Categorical	Yes
Purchase price	Quantitative	—
Supplier lead time	Quantitative	—
Minimum Order Quantity (MOQ)	Quantitative	—
Catalog sale price	Quantitative	—
Actual sale price	Quantitative	—
Product end date	Quantitative. A date is a point in time, and we can find the time between two dates.	—
Margin	Quantitative	—
Past sales (52 weeks, 4 weeks, 1 week) in quantity and value	Quantitative (these are really six variables)	—
Current stock in quantity and value	Quantitative (two variables)	—
Average stock coverage	Quantitative	—